

Verification Report: Evaluating the Multi-Agent Realist Synthesis Framework Against the Richmond et al. (2020) Human Benchmark

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1. Background — What Is This Project and Why Does It Exist?

In 2020, a team of five researchers (Richmond, Cooper, Gay, Atiomo, and Patel) published a realist review titled *"The student is key"* in the journal *Medical Education*. Over several months, the team manually read 28 research papers and produced a structured report answering:

"What works, for whom, in what circumstances, and why — when teaching clinical reasoning to medical students?"

Their method is called **Realist Synthesis**. Unlike a standard literature review that simply counts "does it work?", realist synthesis digs into the causal chain behind each finding by breaking it into three components:

- **Context (C):** The starting conditions. *Who is the student?* (e.g., a beginner with low clinical knowledge)
- **Mechanism (M):** The causal engine. *What does the teaching method provide (Resource), and how does the student react internally (Response)?*
- **Outcome (O):** The result. *What happens?* (e.g., improved diagnostic accuracy, or negative learning outcomes)

A single combination of Context → Mechanism → Outcome is called a **CMOC** (Context-Mechanism-Outcome Configuration). Richmond's team produced multiple CMOCs organized around five student types (Contexts 1–5), then synthesized them into a unified **Programme Theory** — a narrative explanation of how and why things work.

The goal of this project is to test whether an AI system, built with multiple specialized agents and a knowledge graph, can perform the same analytical operations that Richmond's human team performed. If the AI's output aligns with the human benchmark, it provides evidence that this technology can assist researchers in conducting systematic reviews — reducing the manual workload while maintaining analytical rigor. The system does not claim to replace human judgment; it claims to externalize and structure the analytical labor.

2. The E01–E47 Ontology — The Shared Vocabulary

To ensure consistency across papers, Richmond's team created a **standardized dictionary of 47 concepts** (labeled E01 through E47). Every finding from every paper must be mapped to one of these 47 concepts. Our AI system uses the same dictionary, hard-coded as a Pydantic schema. The AI cannot

invent new concepts outside this list — this is the primary mechanism for preventing hallucination.

Context — Who is the student? (E01–E06)

Code	Definition
E01	Undergraduate students in medical or healthcare professions education
E02	Students with "low knowledge," low clinical domain-specific knowledge, or inability to use knowledge in reasoning
E03	Students with high clinical domain-specific knowledge
E04	Positive student coping strategies or appropriate self-confidence/self-efficacy
E05	Negative student coping strategies or lacking self-confidence/self-efficacy
E06	Students with different levels of knowledge within a group

Intervention — What teaching method is used? (E07–E18)

Code	Definition
E07	An expert's reasoning processes are explicitly revealed and discussed
E08	Instructed to use analytical reasoning alone
E09	Teaching resources that allow students to make mistakes
E10	Real-life scenarios, including simulation and simulated patients
E11	Real cases
E12	Strategies that promote knowledge retention
E13	Accurate and timely feedback
E14	Feedback is absent, incomplete, or contains errors
E15	Explicit and clear explanation of expert's reasoning
E16	Passive observation of experts without explanation
E17	Listen to near-peer think aloud their reasoning with prompts and examples
E18	Instructing to use both non-analytical and analytical reasoning

Mechanism Resource — What does the intervention provide? (E19)

Code	Definition
E19	Multiple relevant resources

Mechanism Response — How does the student react internally? (E20–E45)

Code	Definition
E20	Understanding
E23	Frustrated
E24	Rely on non-analytical reasoning
E26	Grateful for the learning experience
E27	Build understanding
E31	Pressure that their decision making could have a real impact
E32	Fear
E33	Stress
E34	Pressure to perform
E35	Cognitive load is increased
E39	Build upon what they already know
E42	Develop understanding of successes/failures and generate improvement plans
E45	Confusion

Outcome — What is the measurable result? (E21–E47)

Code	Definition
E21	Insight into the reasoning process when diagnosing and managing patients
E22	Positive learning experience
E25	High diagnostic accuracy
E28	Positive impact on learning
E29	More complete illness scripts
E30	More accurate non-analytical reasoning

E36	Poor illness script development
E37	Faulty future non-analytical reasoning
E38	Negative learning outcomes
E40	Increased learning
E41	Further engagement
E43	Complete illness scripts
E44	Successful non-analytical reasoning in the future
E46	Increase in learning gain or diagnostic accuracy
E47	Decrease in learning gain or diagnostic accuracy

3. System Architecture — How the AI Pipeline Works

The system has two major components: a **10-Agent Pipeline** (built with LangGraph) and a **Knowledge Graph** (built with Microsoft GraphRAG). They serve different but complementary purposes.

3.1 The 10-Agent Pipeline (LangGraph)

LangGraph is an open-source framework that models workflows as directed graphs — a series of nodes (agents) connected by edges (transitions). Each agent is a Python function that receives a shared state, performs one specific task, and returns its results. The framework automatically passes the updated state to the next agent.

Why this architecture: Realist synthesis is inherently a sequential process — you cannot extract CMOCs before screening papers, and you cannot write a Programme Theory before extracting CMOCs. A directed graph enforces this ordering while keeping each step isolated and testable.

Step	Agent	Input	Output
0	Protocol Agent	Review question	Initial Programme Theory (IPT)
1	Deduplication Agent	PDF files	Deduplicated study registry with stable IDs
2a	Screening Agent	Titles + abstracts	Include/exclude/uncertain decisions
—	HITL Checkpoint 1	Uncertain cases	Human adjudication
2b	Full-Text Agent	Full paper text	Final included list

3	CMOC Extractor	Full paper text + E01–E47 schema	Structured C-M-O data
—	HITL Checkpoint 2	Extracted CMOCs	Human verification
4	Contradiction Agent	All CMOCs	Contradictions identified
—	HITL Checkpoint 3	Contradictions	Human resolution
5	Theory Synthesizer	Validated CMOCs + contradictions	Narrative Programme Theory
—	HITL Checkpoint 4	Draft theory	Human sign-off
6	Reporting Agent	All pipeline data	PRISMA report, evidence tables, audit trail

HITL (Human-In-The-Loop): At four points, the pipeline pauses and waits for human review. In the current pilot, these checkpoints auto-approved to allow the pipeline to complete. In a production setting, a researcher would review and approve at each checkpoint.

3.2 The Knowledge Graph (Microsoft GraphRAG)

GraphRAG processes the same papers through a separate pipeline:

1. **Text chunking:** Each paper is split into overlapping segments.
2. **Entity extraction:** An LLM reads each segment and identifies concepts (entities) and their relationships.
3. **Graph construction:** All entities and relationships are assembled into a single large graph.
4. **Community detection (Leiden algorithm):** The graph is partitioned into clusters of closely related concepts. Each cluster represents a "topic" or "theme" that emerges from the data.
5. **Community reports:** An LLM summarizes each cluster into a human-readable report.

Why this matters: The Leiden communities represent what Richmond's team discovered through months of reading — the recurring themes and concepts across the literature. If the AI's communities match Richmond's themes, it suggests the knowledge graph accurately captures the conceptual structure of the corpus.

4. Verification — Step-by-Step Comparison Against Richmond

This section follows the verification protocol from Professor Wei Zheng's "Plan for Model Verification" (Part A, Steps 1–8).

Step 1: Search & Screening Alignment

Richmond's benchmark: 28 papers included after searching 4 databases and applying inclusion/exclusion criteria.

Our setup: We fed the same 28 papers directly into the system (since we are using Richmond's corpus as the benchmark, not performing an independent search). The AI then screened these papers.

Results:

Stage	Richmond	AI System
Papers input	28 (from 4 databases)	28 (same papers, provided as input)
After deduplication	28	28
After title/abstract screening	28 included	10 included, 16 excluded, 2 uncertain
After full-text review	28 included	12 included

Assessment: There is a significant divergence. The 28 papers we provided are Richmond's *final included set* — they had already passed all human screening. The AI should ideally have retained all 28. The fact that it excluded 16 suggests the Screening Agent applied its criteria too aggressively. This is a known issue with LLM-based screening: the AI tends toward conservative exclusion when abstracts are ambiguous or incomplete (due to PDF extraction artifacts).

Impact: Downstream outputs (CMOC extraction, Programme Theory) are based on only 12 papers instead of 28.

Step 2: CMO Extraction Fidelity

Richmond's benchmark: For each included paper, the human team extracted C-M-O components using the E01–E47 codes.

Our test: We ran extraction on one paper (Linn et al., 2012) as a pilot.

Result — direct comparison for Linn (2012):

C-M-O Field	AI Extracted	Code	What Richmond Coded	Match?
Context	"students with 'low knowledge,' low clinical domain-specific knowledge, or an inability to use knowledge in a reasoning situation"	E02	Context 1 = E02 (low knowledge students)	✓
Intervention	"teaching resources that allow them to make mistakes"	E09	Clinical teaching framework where students practice	✓

Mechanism Resource	"multiple relevant resources"	E19	"multiple relevant resources" (same phrase)	✓
Mechanism Response	"build understanding"	E27	Building student understanding	✓
Outcome	"positive impact on learning"	E28	Positive learning outcomes	✓

Assessment: All five fields match Richmond's coding. Zero hallucination — every label is a valid E-code from the ontology.

Important caveat: This is one paper out of 28. A sample of $n = 1$ cannot establish statistical reliability. To properly assess extraction fidelity, extraction must be run on all included papers and compared field-by-field against Richmond's full coding. This pilot demonstrates that the extraction *mechanism works correctly*, not that it will work correctly on every paper.

Step 3: Knowledge Graph — Entities and Relationships

Richmond's benchmark: Richmond did not produce a knowledge graph. This is a novel contribution of our system that extends beyond what the human team did.

GraphRAG output statistics:

Metric	Value	Interpretation
Documents processed	20 out of 28	71% of Richmond's corpus (8 PDFs failed extraction)
Entities extracted	1,318	Concepts identified across all papers
Relationships extracted	1,672	Connections between concepts
Relationship-to-entity ratio	1.27	Each entity connects to ~1.3 others on average
Community clusters	245	Thematic groupings found by Leiden algorithm

Assessment: The entity types in GraphRAG were configured to match Richmond's C-M-O categories (Context, Intervention, Mechanism_Resource, Mechanism_Response, Outcome). This means the knowledge graph is structured around the same conceptual framework that Richmond used. The 1,318 entities and 1,672 relationships form a rich, interconnected network. However, with only 71% corpus coverage, some themes may be under-represented.

Step 4: Community–Mechanism Alignment

This step checks whether the themes that emerged from GraphRAG's automatic clustering correspond to the themes that Richmond's human team identified through manual reading.

Source of Richmond's themes: Richmond's paper discusses the following topics extensively across multiple sections (Introduction, Results, Discussion). These are not a numbered list in the paper, but rather the dominant subjects that the authors repeatedly analyze and discuss:

#	Topic Richmond discusses at length	GraphRAG Community	Rank	Found?
1	Diagnostic reasoning strategies and their effect on accuracy	"Diagnostic Accuracy and Reasoning Strategies in Medical Education"	8.5	✓
2	Self-explanation as a technique for improving clinical reasoning	"Clinical Reasoning and Self-Explanation in Medical Education"	8.5	✓
3	Simulation and game-based learning for clinical decision-making	"Medialis and Surgical Training Community"	8.5	✓
4	Virtual patient cases and EHR-based education	"Casebook Application and Virtual Patient Education"	8.5	✓
5	Repeated testing and assessment for knowledge retention	"Medical Education Assessment Community"	8.5	✓
6	Cognitive biases affecting diagnostic decisions	"Cognitive Biases and Diagnostic Errors in Medical Education"	8.0	✓
7	Cognitive Forcing Strategies (CFS) as a debiasing tool	"Cognitive Forcing Strategies and Diagnostic Errors"	7.5	✓
8	Structured reflection to improve diagnostic competence	"Structured Reflection in Medical Education"	7.5	✓
9	Schema-based instruction for clinical reasoning	"Schema-Based Instruction in Medical Education"	7.5	✓
10	Combining analytic and non-analytic reasoning approaches	"Combined Reasoning Strategies for Novice Diagnosticians"	7.5	✓
11	Dual-Process Theory as the theoretical foundation	No dedicated community	—	△ Partial

Assessment: 10 out of 11 topics match. Dual-Process Theory is the theoretical lens that underlies the entire review — it is referenced throughout the paper but not as a standalone intervention or context. It makes sense that the algorithm did not cluster it as a separate community because it is a meta-level concept, not a specific educational intervention.

Note on methodology: The 11 topics listed above were identified by reading Richmond's paper and extracting the major subjects of analysis. This is a qualitative comparison — a more rigorous evaluation would involve an independent rater (e.g., a second researcher) identifying Richmond's themes and comparing them to the communities.

Step 5: Programme Theory Comparison

Richmond's Programme Theory is organized around five student Contexts:

- **Context 1 (E02):** Students with low knowledge
- **Context 2 (E03):** Students with high knowledge
- **Context 3 (E04):** Students with positive coping strategies / self-efficacy
- **Context 4 (E05):** Students with negative coping strategies
- **Context 5 (E06):** Mixed-level groups

For each Context, Richmond describes a full CMOC chain (Context → Intervention → Mechanism → Outcome).

AI's Programme Theory covers only **Context 1 (E02)**. Below is a field-by-field comparison for that Context:

Element	Richmond's Statement	AI's Statement	Match?
Target population	Students with low knowledge (Context 1)	"novice learners... who possess 'low knowledge,' limited clinical domain-specific knowledge"	✓
Intervention	Expert demonstration with resources that allow practice	"teaching resources that allow students to make mistakes... clinicians demonstrating clinical consultations"	✓
Mechanism response	Building understanding of reasoning processes	"designed to build understanding by assisting students in comprehending the clinician's thought process"	✓
Outcome	Positive impact on learning and reasoning development	"positive impact on learning, specifically the development of clinical reasoning skills"	✓
Causal logic (C→M→O)	Present and explicit	Present and explicit	✓
Academic writing quality	Publication-ready	Clear, logical, professional tone	✓

What is missing compared to Richmond:

Element	Present in Richmond?	Present in AI?
Context 2 analysis (high knowledge)	✓	✗
Context 3 analysis (positive coping)	✓	✗
Context 4 analysis (negative coping)	✓	✗

Context 5 analysis (mixed groups)	✓	✗
Contradictory findings (e.g., CFS)	✓	✗ "None detected"
Demi-Regularities	✓	✗ "Not yet computed"

Assessment: For Context 1, the AI's theory is substantively correct. But it covers only 1 of 5 Contexts because only 1 CMOC was extracted. This is a limitation of the pilot run, not of the system architecture.

Step 6: Audit Trail and Transparency

RAMESES Transparency Criterion	Richmond	AI System
Search process documented	Yes (Methods section of paper)	Yes (audit_trail.md, entry 2: "Registry built: 28 unique studies")
Screening decisions with rationale	Aggregated counts only	Per-paper decisions: "10 include, 16 exclude, 2 uncertain"
Human review checkpoints	Described generally in Methods	Logged explicitly: 4 named HITL checkpoints with entry numbers
Data extraction logged	Described in Methods	Logged: "Extracted 5 entities, 4 relations from [paper]"

Assessment: The AI system provides a more granular audit trail than Richmond's published paper. Each agent decision is logged individually. This is an advantage for reproducibility.

Step 7: Evidence-Grounded Q&A Demonstration

The Plan for Model Verification (Step 7) specifies testing the system with realist questions such as *"What works, for whom, in what contexts, and why?"*

Status: This step has not been performed yet. To complete it, we would need to:

1. Define a fixed set of realist questions based on Richmond's review question.
2. Run each question through GraphRAG's query engine.
3. Evaluate whether the AI's answers cite specific papers, use C-M-O framing, and align with Richmond's conclusions.

This is a gap in the current evaluation. It should be addressed in the next phase.

Step 8: Boundary Statement — What Is and Is Not Replicated

Following the Plan for Model Verification (Step 8):

What the system successfully reproduced (simulated):

Operation	Evidence
Screening logic	Agent applied inclusion/exclusion criteria and produced reasoned decisions (though over-excluded)
CMO extraction with ontology enforcement	1 CMOC extracted with 5/5 field accuracy, using E01–E47 schema
Knowledge graph construction	1,318 entities, 1,672 relationships from 20 papers
Cross-study pattern detection	245 Leiden communities; 10/11 match Richmond's themes
Programme Theory generation	Context 1 theory structurally and substantively matches Richmond
Decision audit trail	More detailed than Richmond's published methodology

What the system did NOT replace:

Operation	Reason
Final interpretive judgment	Deliberate design choice — humans retain authority per realist methodology
Normative theory selection	The AI generates candidate theories; humans decide which is "best"
Full-scale extraction (all 28 papers)	Not yet executed — pilot ran on 1 paper only
Contradiction resolution	Requires multiple CMOCs; not triggered in pilot
Independent search and retrieval	Papers were pre-supplied; the AI did not perform literature search

5. Summary of Findings

Verification Step	Status	Key Finding
Step 1: Search & Screening	⚠️ Partial	AI retained 12/28 papers (over-excluded). Screening Agent needs recalibration.
Step 2: CMO Extraction	✓ Pilot passed	1 CMOC extracted, 5/5 fields correct. Needs full-scale validation.
Step 3: Knowledge Graph	✓ Built	1,318 entities, 1,672 relationships, 71% corpus coverage.

Step 4: Community Alignment	✓ Strong	10/11 themes match Richmond (91% alignment).
Step 5: Programme Theory	✓ Partial	Context 1 correct. Contexts 2–5 not yet covered.
Step 6: Audit Trail	✓ Strong	More granular than Richmond's published process.
Step 7: Q&A Demonstration	✗ Not done	Needs to be performed in next phase.
Step 8: Boundary Statement	✓ Documented	Clear separation of what is replicated vs. not replaced.

6. Conclusion

The pilot run provides evidence that the system's architecture is **capable of performing the core analytical operations of realist synthesis**: screening, CMO extraction, cross-study pattern detection, and programme theory generation. The single CMOC extracted from Linn (2012) matches Richmond's coding across all five fields, and the Knowledge Graph independently identified 10 of 11 major themes from the literature.

However, the current output is a **proof of concept**, not a complete replication. Three items must be addressed before the system's output can stand alongside Richmond's results:

1. **Run extraction on all included papers** to produce a full set of CMOCs covering Richmond's 5 Contexts.
2. **Recalibrate or bypass screening** for this benchmark test, since the input papers are already Richmond's final included set.
3. **Complete Step 7** (evidence-grounded Q&A) to demonstrate end-to-end query capability.

Once these steps are completed, the system will have produced a full set of comparable outputs for formal evaluation against the human benchmark.

This report follows the verification protocol outlined in "Plan for Model Verification" (Part A, Steps 0–8), provided by Professor Wei Zheng.